Robotics and Embodied Autonomous Systems table (for Prof. Simone Baldi)

Sep.-28	*Lecture* – Introduction to connected automated vehicles *Embodied Lecture –* *Marco Pavone Foundation Models for Autonomous Driving*	
Oct.-5	*Lecture* – Introduction to adaptive switched systems *Embodied Lecture –* *Yann LeCun World Models Enabling the next AI revolution*	
Oct.-12	*Lecture* – Introduction to adaptive autopilots *Embodied Lecture –* *Sergey Levine Robotic Foundation Models*	Assignment 1 is provided
Oct.-19	*Lecture* –Introduction to learning-based energy systems *Embodied Lecture –* *Richard Sutton The OaK Architecture – A Vision of Superintelligence from Experience*	
Oct.-26		Deadline for Assignment 1 Assignment 2 is provided
Nov.-2		
Nov.-9		Deadline for Assignment 2
Nov.-16		
Nov.-23		
Nov.-30		
Dec.-7		
Dec.-14		

General Info

Final grade: (tentative) **20 - 20 - 20%** from assignment 1 - 2 - 3 **40%** from final exam

Attendance will be monitored during all classes.

Instructions for assignment 1 - 2

List of exercises for Assignment 1

Chapter 3: Acrobots, Cart-Poles, and Quadrotors (**Example 3.1 + Example 3.2 + Example 3.5 + Example 3.6 + Section 3.8**) https://underactuated.csail.mit.edu/acrobot.html#section8
It is highly suggested to check the lecture
6.8210 Spring 2024 Lecture 5: Acrobots, Cart-poles, and Quadrotors I
https://www.youtube.com/watch?v=UBPL0IbyJy4&list=PLkx8KyIQkMfU5szP43GlE_S1QGSPQfL9s&index=5

Chapter 4: Simple Models of Walking and Running (**Example 4.3 + Example 4.4 + Example 4.5 + Example 4.6 + Section 4.6**) https://underactuated.csail.mit.edu/simple_legs.html#section6
It is highly suggested to check the lecture
Lecture 15: MIT 6.832 Underactuated Robotics (Spring 2022) | "Simple Models of Walking"
https://www.youtube.com/watch?v=N37FMfOioK0&list=PLkx8KyIQkMfU5szP43GlE_S1QGSPQfL9s&index=13

Chapter 5: Highly-articulated Legged Robots (**Example 5.1 + Example 5.2 + Section 5.8**)
https://underactuated.csail.mit.edu/humanoids.html#section8
It is highly suggested to check the lecture
6.8210 Spring 2024 Lecture 16: Humanoid Robots
https://www.youtube.com/watch?v=cRu4EqBswbk&list=PLkx8KyIQkMfU5szP43GlE_S1QGSPQfL9s&index=16

Chapter 1: Introduction (**Example 1.2 + Example 1.3 + Section 1.7**)
https://manipulation.csail.mit.edu/intro.html#section7
It is highly suggested to check the lecture
6.4210 Fall 2023 Lecture 1: Intro
https://www.youtube.com/watch?v=v04rn86Dehg&list=PLkx8KyIQkMfWr191lqbN8WfV08j-ui8WX

List of exercises for Assignment 2

Chapter 2: Let's get you a robot (**Example 2.2 + Example 2.3 + Example 2.4 + Example 2.5 + Example 2.6 + Example 2.7 + Example 2.8 + Section 2.7**) https://manipulation.csail.mit.edu/robot.html#section7
It is highly suggested to check the lecture
6.4210 Fall 2023 Lecture 2: Let's get you a robot!
https://www.youtube.com/watch?v=q896_lTh8eA&list=PLkx8KyIQkMfWr191lqbN8WfV08j-ui8WX&index=2

Chapter 3: Basic Pick and Place (E**xample 3.2 + Example 3.3 + Example 3.5 + Example 3.6 + Example 3.7 + Example 3.8 + Example 3.10 + Example 3.11 + Example 3.12 + Example 3.13 + Example 3.14 + Section 3.11**) https://manipulation.csail.mit.edu/pick.html#section11
It is highly suggested to check the three lectures

6.4210 Fall 2023 Lecture 3: Basic Pick and Place (Pt. 1)
https://www.youtube.com/watch?v=0-34RZJxyww&list=PLkx8KyIQkMfWr191lqbN8WfV08j-ui8WX&index=3
6.4210 Fall 2023 Lecture 4: Basic Pick and Place (Pt. 2)
https://www.youtube.com/watch?v=1mkzXp9_QYY&list=PLkx8KyIQkMfWr191lqbN8WfV08j-ui8WX&index=4
6.4210 Fall 2023 Lecture 5: Basic Pick and Place (Pt. 3)
https://www.youtube.com/watch?v=YaQrC_Zm8qg&list=PLkx8KyIQkMfWr191lqbN8WfV08j-ui8WX&index=5

Chapter 8: Manipulator Control (**Example 8.1 + Example 8.2 + Example 8.3 + Example 8.4 + Example 8.5 + Section 8.6**) https://manipulation.csail.mit.edu/force.html#section6
It is highly suggested to check the two lectures
6.4210 Fall 2023 Lecture 14: Force Control
https://www.youtube.com/watch?v=JtJyesuYvb4&list=PLkx8KyIQkMfWr191lqbN8WfV08j-ui8WX&index=14
6.4210 Fall 2023 Lecture 15: Manipulator Control
https://www.youtube.com/watch?v=9GB6kpQyMtM&list=PLkx8KyIQkMfWr191lqbN8WfV08j-ui8WX&index=16

List of exercises for Assignment 3

To be decided

How to provide your solution

For exercises marked as *Example X.X*: the code is already provided and the exercise is already solved. The student is asked to run the code and comment on the results based on the theoretical concepts of the lecture. When applicable, a sensitivity study based on different parameters is appreciated.

For exercises marked as *Section X.X*: all exercises of the section should possibly be solved. When applicable, the code (or the most important part of the code) should be included in the report.

The student is asked to provide a report (PDF format) and the Matlab (or Python) code. The report should contain the steps used to solve the problems, and the relevant plots from the simulations. Report and code can be sent to simonebaldi@seu.edu.cn the day of the deadline, before 10pm

Each assignment must be solved individually. Copying solutions from other students will be penalized.